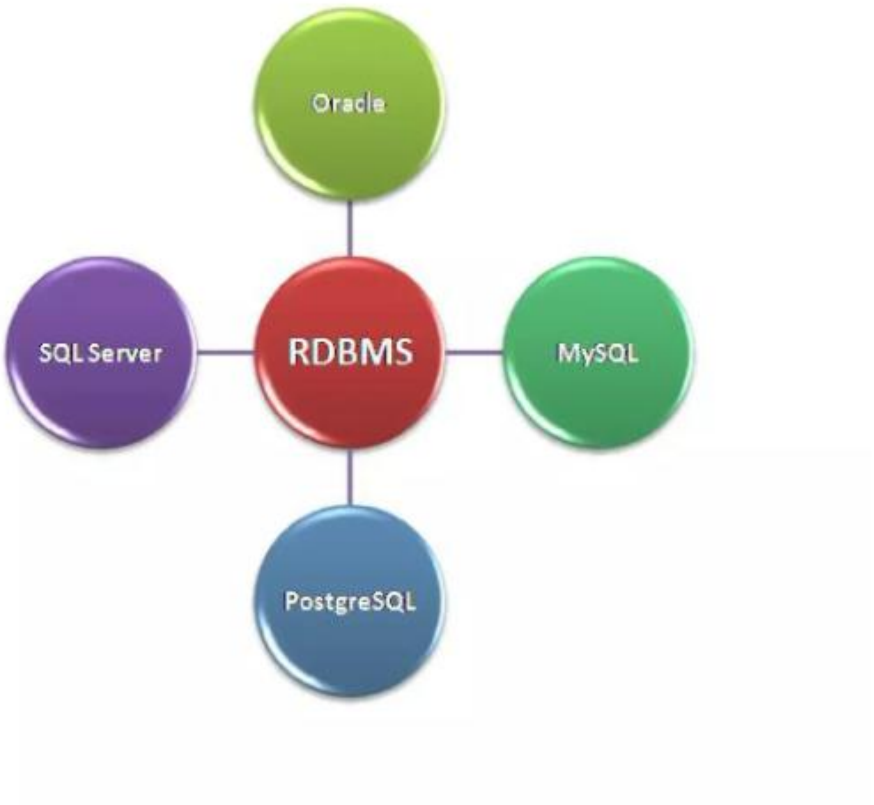
RDBMS Core Concepts-47

Relational Database Management System (RDBMS) is software that organizes data into related tables. Each table consists of rows and columns, making it easier to store, manage, and query structured data. RDBMS is widely used in banking, e-commerce, government, healthcare, and education due to its reliability and structured approach. Below are the detailed core concepts.

We have different types of RDBMS



1. Table – Relational

- A table is the fundamental structure of RDBMS, representing data in rows and columns.
- Columns represent attributes or fields.
- Rows represent records or instances.
- Example: A "Students" table with columns like Student_ID, Name, Age, and Course.

2. Rows/Records

- Each row in a table represents one record.
- Records are unique entries that hold values for each column.
- Example: A row in the "Students" table may be (101, Ravi, 21, BSc).

3. Keys (Primary, Foreign, Candidate)

- Primary Key: A unique identifier for each record in a table. Example: Student_ID.
- Foreign Key: A column that creates a link between two tables. Example: Course_ID in Students table referencing Course_ID in Courses table.
- Candidate Key: A set of attributes that can uniquely identify a record but only one becomes the Primary Key.

RDBMS Core Concepts
1. Table - Relational
2.Rows/Records
3.Primary Key, Foreign Key, Candiate Key
4.Relationships 1 to 1, 1 to M, M to M.
5. ACID Properties -- Atomicity Consistency,
Isolation Durability
6. SQL -- DDL, DML,DQL, DCL,TCL

DDL---Create , ALter, Drop
DML-- Create, Alter Delete
DQL -- Select, Join
DCL -- Grant, Revoke
TCl-- Commit, Rollback, SavePoint

4. Relationships (1 to 1, 1 to Many, Many to Many)

- One-to-One (1:1): One record in a table relates to one record in another table. Example: Student and Student_Profile.
- One-to-Many (1:M): One record in a table relates to many records in another table. Example: One Course has many Students.
- Many-to-Many (M:M): Many records in one table relate to many in another table. Example: Students enrolled in multiple Courses.

5. ACID Properties

- Atomicity: Ensures transactions are all-or-nothing. Example: Money transfer either completes or fails entirely.
- Consistency: Guarantees data validity before and after a transaction. Example: Total balance remains the same across accounts.

- Isolation: Transactions occur independently without interference. Example: Two people booking the same ticket do not overwrite each other.
- Durability: Once a transaction is committed, it remains saved even after a crash.

6. SQL (DDL, DML, DQL, DCL, TCL)
   Structured Query Language (SQL) is used to interact with RDBMS. It is divided into categories:

- DDL (Data Definition Language)
  • Defines database structures.
  • Commands: CREATE, ALTER, DROP.
  • Example: CREATE TABLE Students (ID INT, Name VARCHAR(50));
- DML (Data Manipulation Language)
  • Modifies data inside tables.
  • Commands: INSERT, UPDATE, DELETE.
  • Example: INSERT INTO Students VALUES (101, 'Ravi', 21, 'BSc');
- DQL (Data Query Language)
  • Fetches data from the database.
  • Commands: SELECT, JOIN.
  • Example: SELECT * FROM Students;
- DCL (Data Control Language)
  • Manages permissions and access control.
  • Commands: GRANT, REVOKE.
  • Example: GRANT SELECT ON Students TO user1;
- TCL (Transaction Control Language)
  • Controls transactions in a database.
  • Commands: COMMIT, ROLLBACK, SAVEPOINT.
  • Example:
  BEGIN;
  UPDATE Students SET Age=22 WHERE ID=101;
  COMMIT;

RDBMS core concepts such as tables, keys, relationships, ACID properties, and SQL operations form the backbone of modern databases. They ensure data accuracy, security, and reliability. Mastering these concepts is essential for working with structured data in real-world applications.